

Econ 646 Spring 2013
Homework 3

This homework is due on March 4th, at the beginning of the lecture.

1. *No capital model* No-capital version of the RBC model is

$$\max_{\{C_t, N_t\}_{t=0}^{\infty}} E_0 \left[\sum_{t=0}^{\infty} \beta^t (\log C_t + \theta \log (1 - N_t)) \right]$$

subject to

$$C_t = A_t N_t$$

where A_t is a stationary random variable, representing a technology shock.

- (a) Derive the first order conditions.
 - (b) How does N_t respond to a change in A_t ?
 - (c) Explain your answer in (b). (HINT: Notice that the utility function is a King-Plosser-Rebelo type.)
2. Replicate Figure 4 of Leeper, Sims, and Zha (1994). You can use industrial production index, instead of monthly GDP
3. Discuss your results on the following points. What do they tell about the monetary policy effects? What are the possible issues of your interpretation of the results?